

① Problem $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = |x-2|$

(a) Is f convex?

Yes $|x-2|$ is convex with unique minimizer

at $x^* = 2$. A function $f: \mathbb{R} \rightarrow \mathbb{R}$ is convex if for any

(b) $x_1, x_2 \in \mathbb{R}$ and any $\lambda \in (0, 1)$

$$f(\lambda x_1 + (1-\lambda)x_2) \leq \lambda f(x_1) + (1-\lambda)f(x_2)$$

Intuitively the line segment between any two points on the graph f lies above the graph itself.

check for $f(x) = |x-2|$ we can think of this as a shifted absolute value function $f(x) = |x-2| = g(x-2)$ where $g(x) = |x|$

We already know that $|x|$ is convex, A shift by 2 doesn't affect convexity - it just moves the graph horizontally.

so $f(x) = |x-2|$ is convex.

(b) Bounded Gradients :

$$f'(x) = \begin{cases} -1 & x < 2 \\ 0 & x = 2 \\ 1 & x > 2 \end{cases}$$

As the derivative with $f'(2) = 0$

Taking $y=2$ and $x=2+\epsilon$ gives $|f'(x) - f'(y)| = |1-0| = 1 \leq L\epsilon$

which fails for arbitrary small $\epsilon > 0$. Hence no global L

(c) Gradient Descent from $x_0 = 1.05$, $\eta = 0.1$

Update $x_{i+1} = x_i - \eta f'(x_i)$

For $x_i < 2$; $f'(x) = -1 \Rightarrow x_{i+1} = x_i + 0.1$

For $x > 2$: $f'(x) = 1 \Rightarrow x_{i+1} = x_i - 0.1$

From $x_0 = 1.05$, $x_t = 1.05 + 0.1t$ until $x_{10} = 2.05$, then it flips $x_{11} = 1.95$ and the sequence oscillates.

It does not converge to the minimizer $x^* = 2$. Function values stall at $f(1.95) = f(2.05) = 0.05 > 0$

Thus, with constant step 0.1, Gradient descent does not converge to the global minimum.

② Regression Problems

① Rescale the responses by a constant $y'_i = \alpha y_i$

For a simple linear regression with intercept,

$$b_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \quad b_0 = \bar{y} - b_1 \bar{x}$$

under $y'_i = \alpha y_i$

$$\sum (x_i - \bar{x})(y'_i - \bar{y}') = \alpha \sum (x_i - \bar{x})(y_i - \bar{y})$$

$$\text{Hence } b'_1 = \alpha b_1, \quad b'_0 = \bar{y}' - b'_1 \bar{x} \Rightarrow \alpha(\bar{y} - b_1 \bar{x}) \Rightarrow \alpha b_0$$

Result: both coefficients scale by α :

$$b'_0 = \alpha b_0, \quad b'_1 = \alpha b_1$$

2-(b) what happens if x_i 's scaled by α

Recall for simple linear regression $y = b_0 + b_1 x$

$$b_1 = \frac{\text{Cov}(x, y)}{\text{Var}(x)}, \quad b_0 = \bar{y} - b_1 \bar{x}$$

if inputs are scaled $x_i' = \alpha x_i$

New mean $\bar{x}_i' = \alpha \bar{x}_i$

New variance $\text{Var}(\bar{x}_i') = \alpha^2 (\text{Var}(x))$

New Covariance: $\text{Cov}(x', y) \Rightarrow \text{Cov}(\alpha x, y)$
 $\Rightarrow \alpha (\text{Cov}(x, y))$

So

$$b_1' = \frac{\text{Cov}(x', y)}{\text{Var}(x')} \Rightarrow \frac{\alpha (\text{Cov}(x, y))}{\alpha^2 \text{Var}(x)} \Rightarrow \frac{1}{\alpha} b_1$$

and

$$b_0' = \bar{y} - b_1' \bar{x}' \Rightarrow \bar{y} - \frac{1}{\alpha} b_1 (\alpha \bar{x}) \Rightarrow \bar{y} - b_1 \bar{x} = b_0$$

Result: Intercept stays the same:

$$b_0' = b_0$$

Slope rescales inversely with α :

$$b_1' = \frac{b_1}{\alpha}$$

2. (C)

(i) Can we use linear regression to learn $y = \alpha x_1 + \beta x_2$

yes. A general multiple linear regression model is

$$\hat{y} = b_0 + b_1 x_1 + b_2 x_2$$

As our relation ship is $y = \alpha x_1 + \beta x_2$. This is exactly of that form - just with $b_0 = 0$. So linear regression can represent it perfectly

(ii) Can we learn $\alpha x_1^2 + \beta x_2^2$.

yes As these inputs are non linear, but linear regression works on linear data; we need to transform the input features. ex: $z_1 = x_1^2$, $z_2 = x_2^2$ then our model becomes $\alpha z_1 + \beta z_2$ which is linear

(iii) Can we learn $y = 2^\alpha x_1^\beta$

NO. Linear regression cannot learn $y = 2^\alpha x_1^\beta$ because it is non linear in parameters α, β .

$$(3) \quad S = (1, 3), (2, 6), (3, 9)$$

(a) Each point satisfies $y = 3x$.

So all points lie exactly on the line through the origin with slope 3. Geometrically the data is 1-dimensional.

In R^2 , the ~~max~~ maximum number of PC is 2. But here all variance lies along a single line. The covariance matrix will have rank = 1. So only 1 nonzero principal component

Computing the first Principal Component

"Direction of maximum variance is along the vector (1, 3)

Normalize to get a unit vector

$$v = \frac{1}{\sqrt{1^2 + 3^2}} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

(b) The best rank-1 approximation of a matrix A from its SVD $A = U \Sigma V^T$ is given by

keeping only the largest singular value and its singular vectors.

$$A_1 = \sigma_1 u_1 v_1^T$$

where σ_1 = largest singular value

u_1 = first (left) singular vector

v_1 = first (right) singular vector.

3-c If $A \in \mathbb{R}^{3 \times 3}$, then the scores of the data projected onto the first k principal components are

$$Z_k = AV_k = U_k \Sigma_k$$

This only uses U and Σ (~~since~~ AV_k)

$$\text{Since } AV_k = (U \Sigma V^T)(V_k)$$

$$\Rightarrow U \Sigma \begin{bmatrix} I_k \\ 0 \end{bmatrix}$$

$$\Rightarrow U_k \Sigma_k$$

④ PAC learning for Origin Centered Concentric Circles

(a) Instance space: points in $\mathbb{R}^2$

Concept class $C = \{C_{r_1, r_2} : r_1 < r_2\}$ where

$$C_{r_1, r_2}(x) = \begin{cases} 1 & \text{if } r_1 \leq \|x\| \leq r_2 \\ 0 & \text{otherwise} \end{cases}$$

So the true concept is some pair (r_1, r_2) points between two radii are positive, everything else is negative.

Steps:

- ① Convert each sample point to its distance from the origin
- ② Among all positive samples, take the smallest and largest radius
- ③ Predict +ve for any new point whose distance from the origin lies between those two radii otherwise negative
- ④ This hypothesis is guaranteed to PAC-learn the

concept class $\mathcal{C}$ because it is equivalent to learning 1D intervals, which have VC dimension 2

4(b) The bad events are when the sample doesn't include points close enough to the true inner or outer boundary of the positive region. Missing these points causes large classification error.

4(c) For any concept class $\mathcal{C}$ of VC-dimension d

$$m \geq \frac{1}{\epsilon} \left(d \ln \frac{2}{\epsilon} + \ln \frac{2}{\delta} \right)$$

when $d=2$

$$m = O \left(\frac{1}{\epsilon} \left(\ln \frac{1}{\epsilon} + \ln \frac{1}{\delta} \right) \right)$$

6

(a) If the data are symmetric above and below, the mean of all the points lie approximately at the origin

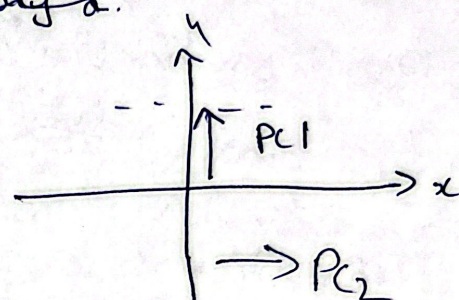
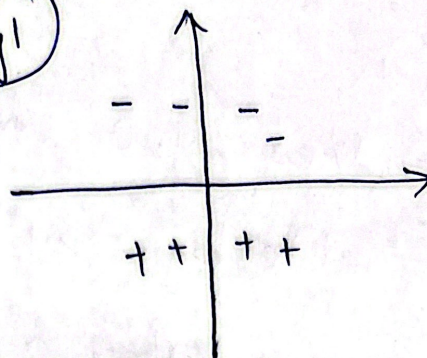
Most variation is vertical (between +ves and -ves along y-axis)

~~If the points are spread horizontally.~~

PC1 will be vertical along y-axis

PC2 will be along x-axis.

Fig 1



Mean of the data lies in Quadrant 1, not at the origin.

Fig 2

The main variation runs through roughly vertically because the points are separated in y-direction.

PC1 $\approx$ Vertical axis (y-direction)

Some small variation that is secondary which is horizontal to x, but smaller

PC2 $\approx$ horizontal axis (x-direction).

6-(b)

Yes, you can have a data set that is linearly separable by some line in $\mathbb{R}^2$ but becomes not linearly separable after projecting on to either principal components.

Consider the Fig 1 example where all the positives lie under the x-axis and -ve samples lie above the x-axis. These two are linearly separable.

Once they are projected onto PC1 (y-axis), they overlap. Their 1-D projections are identical, so no threshold can separate them.